

Comparison of Chinese medicine additives for antibiotic resistance risks in the Wuding chickens under high stocking density: Advantages and drawbacks

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ABSTRACT

Traditional Chinese medicines (TCMs) with anti-pathogenic properties have been increasingly utilized in animal husbandry as substitutes for antibiotics. However, lingering uncertainties persist regarding the mobility, bacterial hosts, and potential human hazards associated with antibiotic resistance genes (ARGs) in animal husbandry when TCMs are introduced. In this study, cecal digesta from Wuding chickens subjected to varying stocking densities (moderate and high) and treated with different TCMs (dihydroartemisinin, berberine hydrochloride, and oligomeric proanthocyanidins) were collected and subjected to metagenomics-based analysis. A total of 280 ARG subtypes were identified, with multidrug and aminoglycoside resistance genes being the most diverse and abundant. The primary antibiotic resistance bacteria identified belonged to the phyla of Firmicutes (38.68%), Bacteroidetes (22.74%), and Proteobacteria (12.71%); however, some pathogenic antibiotic-resistant bacteria (PARB) were also identified across different samples. Surprisingly, high stocking density was not observed to propagate ARGs, which could be due to the restricted usage of antibiotics. Furthermore, higher proportions of mobile genetic elements and some multidrug-resistant PARB (~4 times) were observed in high-stocking-density chickens fed berberine hydrochloride than in high-stocking-density chickens fed other TCMs, which could facilitate the horizontal gene transfer (HGT) of ARGs and pose potential human health risks. In contrast, dihydroartemisinin reduced the total abundance of ARGs and the resistome risks to human health (~10% decrease compared to control). Moreover, rank I ARGs with higher human health risk and the risk index based on ARG risk also ranked lowest ($p < 0.05$) in high-stocking-density chickens fed dihydroartemisinin. The reduction effect of dihydroartemisinin was achieved by regulating the cecal microbiome of broilers and inhibiting HGT (reducing the contribution from 17.7% to 4.0%). This study provides metagenomic evidence of the development of ARGs in broiler breeding environments. The findings highlight the distinct characteristics of TCMs as feed additives on regulating the antibiotic resistome during broiler farming and demonstrate dihydroartemisinin as an effective additive complying with the environmental safety.

1. Introduction

Antibiotic resistance genes (ARGs), widely distributed in environmental media, have been defined as a new type of environmental

pollutant (Pruden et al., 2006; Sanderson et al., 2016). ARGs can be transferred worldwide, resulting in an increase in the incidence and mortality of related infections caused by ARG-carrying pathogens (Amarasiri et al., 2020). Currently, the transfer and transmission of

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ARGs represent a challenging issue in global public security (Founou et al., 2016). Animal husbandry has been considered an important source of ARG pollution in the environment (Pereira et al., 2021). ARGs excreted in animal feces can spread into soil and crops, enter the food chain, and ultimately migrate to the human body, posing a significant threat to human health (Rothrock Jr et al., 2021; Zhao et al., 2021a). Consequently, a comprehensive understanding of the fate of ARGs in various animal husbandry systems is imperative to control their potential health risks.

Broiler farming plays a significant role in meat production and the overall food industry (Mottet and Tempio, 2017). Stocking density (the number of broilers in a given area of poultry house), a key factor in broiler farming, influences growth performance, muscle quality, stress translation, and immune health of broilers (Sun et al., 2021; Thema et al., 2022). While high stocking density reduces costs and manifests the main characteristics of modern animal production system technology (Li et al., 2021), it adversely affects animal health and welfare (Abudabos et al., 2013). Overcrowding makes broilers susceptible to various diseases and greatly increases the concentration of harmful gases (H_2S , NH_3 , CO_2 , and CH_4) in the house, resulting in reduced production performance (Zhang et al., 2011; Sugiharto, 2022). Previous studies on the effects of stocking density on animal husbandry have mainly attributed the development of antibiotic resistance to greater antimicrobial use during intensive farming (Rodrigues da Costa and Diana, 2022). However, a direct quantitative relationship between stocking density and antibiotic resistance remains unestablished. This highlights the need for further research and careful monitoring of ARGs in intensive animal farms.

The use of antibiotics as growth promoters in broiler farming has been banned in multiple countries, and their therapeutic use stringently regulated (Shao et al., 2021). To identify more effective options to control the disease outbreaks, especially under high-density breeding modes, an increasing number of researchers are focusing on the potential of traditional Chinese medicines (TCMs) in animal husbandry as a food additive and antibiotic replacement (Gong et al., 2014). TCMs reportedly can improve broiler production performance (Song et al., 2023; Zhou et al., 2023). More importantly, TCMs and their active ingredients have antimicrobial activity on antibiotic-resistant bacteria (ARB) (Li et al., 2022; Xue et al., 2023). TCMs can also eliminate ARB-resistant plasmids (Liu et al., 2022), thereby inhibiting horizontal genetic transfer (HGT), an important mechanism contributing to ARG risk (Su et al., 2020). However, the potential of TCMs to address antibiotic resistance in animal husbandry and its underlying mechanisms require further investigation.

Antibiotic alternatives, such as Berberine, artemisinin, and proanthocyanidins, are typical TCMs feed additives (Xu et al., 2020; Guo et al., 2020). Berberine (a quaternary ammonium alkaloid) isolated from Huanglian, is the main active ingredient responsible for Huanglian's antibacterial activity (Wang et al., 2019). Artemisinin is a sesquiterpene lactone endoperoxide isolated from the traditional Chinese herbal medicine, *Artemisia annua* (Shahrajabian et al., 2020). Proanthocyanidins (polyphenolic compounds) are widely distributed in TCMs plants (Qi et al., 2023). This study employed metagenomics sequencing to investigate the gut microbiota in a representative Chinese local chicken breed under moderate and high stocking with the above three TCMs additions. Based on a comparison of ARGs, mobile genetic elements (MGEs), and pathogenic antibiotic-resistant bacteria (PARB), the objectives of this study were to clarify the following issues: (i) Will high-density broiler breeding promote the development of antibiotic resistance? (ii) Can the addition of TCMs facilitate the removal of ARGs in broilers, and which TCMs is more efficient? (iii) How does TCMs reduce the prevalence of ARGs in broilers? The results of this study could contribute to exploring strategies to combat antibiotic resistance and guide high-density broiler farming practices.

2. Materials and methods

2.1. Broiler design

All animal experiments and procedures were approved by the Animal Care and Use Committee of the Yunnan Agricultural University (No. 202210012). The stocking density groups were set according to the Cai et al. (2019) and Li et al. (2019). In the present study, 456 19-week-old Wuding broilers (a famous breed in the China) with similar body weight were randomly allocated to five groups (6 replicates per group). Broilers at a moderate stocking density (12 birds/m^2) were used as negative controls (moderate stocking treatment, MT group). Four high stocking density groups (16 birds/m^2) were fed the basal diet (high stocking treatment, HT group), or the basal diet supplemented with 600 mg/kg berberine hydrochloride (HB group), 250 mg/kg grape oligomeric proanthocyanidins (HG group), and 80 mg/kg dihydroartemisinin (HD group). Chickens were fed a basal diet (Table S1) according to NY/T 3645–2020. Clean drinking water was available to all chickens during the experiment. The experiment lasted for 10 weeks, and 12 chickens from each group (2 broilers/replicate) were randomly selected after overnight feed withdrawal. Their cecal digesta were harvested and stored at -70°C .

2.2. Metagenomic sequencing, and metagenomic assembly

The cecum microbial DNA from Wuding Chicken was used to construct the metagenomic libraries using the NEB Next Ultra DNA Library Prep Kit (USA). The sequencing platform used was the Illumina NovaSeq 6000. Low-quality reads were eliminated using the NGS QC Toolkit (Patel and Jain, 2012). Kraken2 was used to analyze the microbial community composition (Wood et al., 2019). Then, the terms belonging to the bacterial domain were retained for further analyses according to annotation results. MEGAHIT and MetaGeneMark were used to assemble the contigs and predict potential functional genes (Li et al., 2015; Gemayel et al., 2022). The abundance of predicted genes was determined using BMap (Bushnell, 2014).

2.3. Gene and host taxonomy annotation

To identify the ARGs and MGEs, function genes were compared to the SARG v3.0-M and MobileGeneticElement database, respectively, using BLASTX with a $e\text{-value} \leq 10^{-10}$, similarity $\geq 80\%$, and query coverage $\geq 70\%$ (Yin et al., 2022; Pärnänen et al., 2018). Contigs carrying both ARGs and MGEs were identified as mobile ARG-carrying contigs (ACCs). The ACCs were further aligned to the NCBI-NR database for taxonomic classification to determine the host taxonomy of the ARGs. Pathogens in the cecal bacteria and ARG hosts were screened based on pathogen lists from the MBPD database (Yang et al., 2023a). Subsequently, ESKAPE (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* spp.) and World Health Organization (WHO) priority pathogens that posed more risks for antibiotic therapy were selected based on the list by Yang et al. (2023b).

2.4. Evaluation of resistome risk

To evaluate the resistome risk of the analyzed samples, arg_ranker v3.0 was applied to a metagenomic dataset (Zhang et al., 2021). In this framework, ARGs were divided into four ranks representing their risk levels based on gene mobility, host pathogenicity, and human-associated enrichment. Rank I consisted ARGs with the highest risk, carried by MGEs and hosted by ESKAPE pathogens. Ranks II and III included ARGs carried by MGEs, while Rank IV covered other ARGs enriched in human-associated environments. The resistome risk index was calculated using the arg_ranker results, considering the rank percentage, rank risk code, and total abundance of each rank (Sang et al.,

2023).

2.5. Statistical analysis

Sequence analysis was carried out using the R v4.2.2 and visualized by the “ggplot2” package. The Bray-Curtis distance of antibiotic resistome between samples was calculated (“vegan” package), and principal coordinates analysis (PCoA) (“ape” package) and PERMANOVA (“vegan” package) were further performed to evaluate their variations among different treatments. Tukey’s test (“multcomp” package) detected differences in diversity, total abundance, type abundance, and subtype abundance of ARGs and MGEs, Bray-Curtis distance of antibiotic resistome, abundance of resistant mechanisms, proportions of mobile ARGs, abundance of ARG ranks, risk index of ARGs, and abundance of pathogens and ESKAPE, respectively, among different treatments. Venn diagram (“VennDiagram” package) was constructed to recognize the shared ARGs among all studied samples. Subsequently, random forest model (“RandomForest” package) based on these shared ARGs were established to predict the total abundance of ARGs.

To explore the mechanisms underlying antibiotic resistome variations, procrustes analysis (“vegan” package) was first applied to assess the correlations between antibiotic resistome with total cecal bacteria and pathogens in studied samples, respectively. Then, the relationships between ARG diversity with MGE diversity and cecal bacterial richness, as well as the similarity of the antibiotic resistome with the similarity of the mobilome and cecal bacterial composition, were investigated by linear regression. Correlations among the abundance of ARGs, MGE types, and cecal bacterial phyla were further analyzed by Spearman correlation (“psych” package). Finally, variance partitioning analysis was executed to explore a relative importance of mobile genetic elements and cecal bacteria to antibiotic resistome variations (“vegan” package).

3. Results

3.1. Variations in antibiotic resistome

Table S2 presents basic information regarding the sequenced data and assembly. According to ARG annotation, 280 ARG subtypes belonging to 24 types were identified in the metagenomic dataset (Fig. S1). ARG subtypes was observed among the different treatments (range: 265–268), with multidrug resistance genes (MRGs) being the most numerous (66), followed by MLS (43), aminoglycoside (29), tetracycline (28), and vancomycin (28) resistance genes (Fig. S1). No significant differences in variations in ARG numbers among different treatments were observed ($p > 0.05$; Fig. 1a). Conversely, the total abundance of ARGs decreased in the cecal microbiome of Wuding chickens fed dietary TCMs, especially in the HD group ($p < 0.05$, Fig. 1a). In contrast to the diversity of ARGs, aminoglycoside resistance genes were the most abundant (31.00%), followed by tetracycline (23.58%), MLS (23.37%), and beta-lactam (13.91%) resistance genes (Fig. S2). However, highly diverse multidrug and vancomycin resistance genes accounted for only 0.65% and 0.59% of the total ARGs abundance, respectively (Fig. S2).

PCoA revealed the effects of dietary TCMs on the antibiotic resistome of Wuding chickens, samples from different treatments were separately clustered along the PC2 axis (Fig. 1b). PERMANOVA also demonstrated significant variations in the antibiotic resistome among different treatments ($p < 0.05$). Furthermore, the increase in inter-individual differences in antibiotic resistance between Wuding chickens under high stocking density was observed, and dietary TCMs inhibited this effect, especially in the HD group ($p < 0.05$, Fig. 1c). Additionally, the inhibition of several ARG types by dietary TCMs (Fig. 1d) were observed. Specifically, HG treatment reduced the aminoglycoside and streptomycin resistance gene abundances ($p < 0.05$), whereas the MLS and tetracycline resistance gene abundances significantly declined in the HB and HD groups ($p < 0.05$). At the subtype level, tet(Q) was the most abundant ARG, although its abundance did not vary among the treatments (Fig. S3). Among other main ARG subtypes, the abundances of APH(3')-IIIa, ANT(6)-Ib, and SAT-4 decreased in the HG group; HB

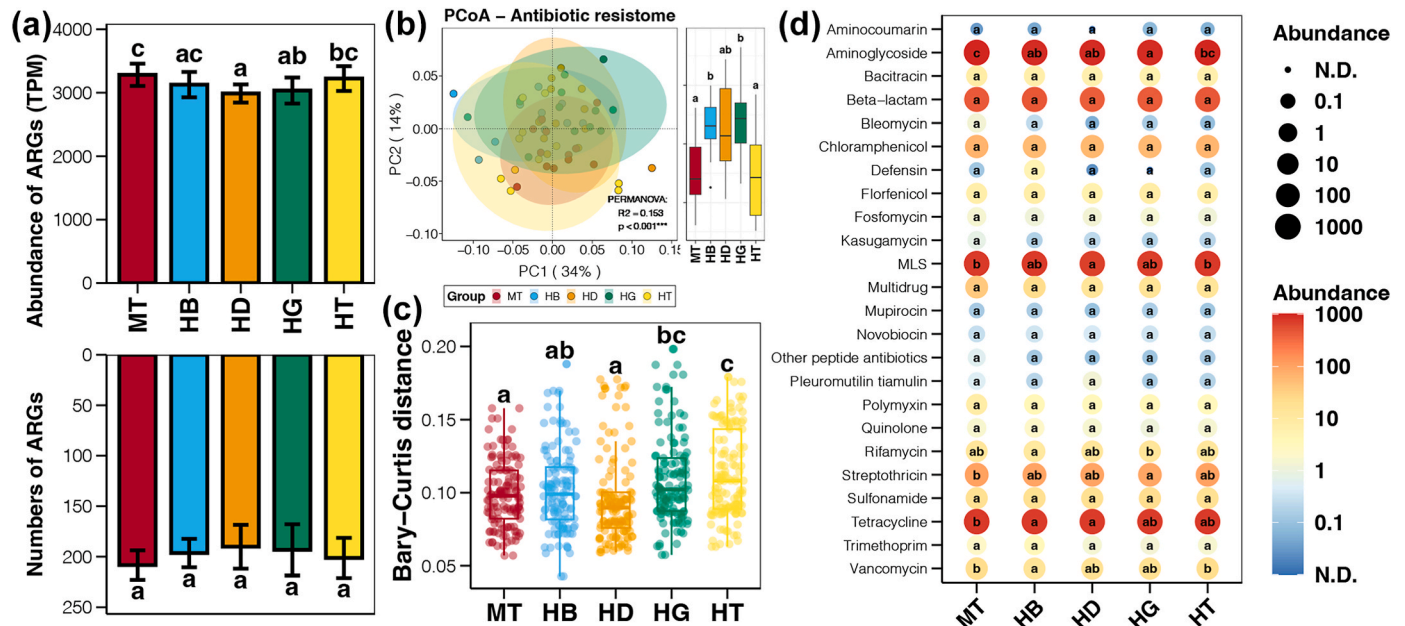


Fig. 1. Variations in antibiotic resistome in cecal microbiome of Wuding chickens with different treatments. (a) Differences in the total abundance and diversity of ARGs among different treatments. (b) PCoA and PERMANOVA based on the Bray-Curtis distance showed the variations in antibiotic resistome among different treatments. (c) Differences in the Bray-Curtis distance between samples with the same treatment. (d) Differences in the abundance of ARG types among different treatments. Different lowercases letters in each subfigure represent significant differences between different treatments (Tukey's HSD test, $p < 0.05$).

treatment inhibited the abundances of *tetW*, *add(6)*, and *lnuc* genes; and the abundance of *ermF* was reduced in the HD group.

3.2. Resistance mechanisms and core ARGs

The SARG database provides information on the mechanisms of action of ARGs against antibiotics. Efflux pump, antibiotic target alteration, and enzymatic inactivation were the most diverse ARG resistance mechanisms found in the dataset (Fig. 2a). The ARG numbers belonging to each resistance mechanism did not vary among the different treatments (Fig. S4). Although these three resistance mechanisms were the most abundant, the proportion of efflux pumps occupied only 4% of the total ARGs, and other resistance mechanisms were lower than 1% (Fig. 2a). Interestingly, the abundances of these three resistance mechanisms varied among the different treatments, being lowest in the HD treatment and highest in the MT treatment (Fig. S5).

A Venn diagram revealed that 89 ARG subtypes were shared among all the studied samples (Fig. S6). The percentages of these shared ARGs nearly accounted for the total ARG abundance (Fig. S7). Among these shared ARGs, the aminoglycoside (20), MLS (17), and tetracycline (17) resistance genes were the most diverse, although they contained only three MRGs (Fig. 2b). A random forest model was constructed based on these shared ARGs to predict the total ARG abundance in the studied samples, and the results showed a 70.9% prediction accuracy (Fig. 2c). *ANT6-Ib*, *ermF*, *aadS*, *aad6*, *OXA-347*, and *tetQ* were recognized as potential key ARGs for the prediction (Fig. S8) and were also ARG subtypes with higher abundance (Fig. S3).

3.3. Variations in mobilome and ARG risk

According to the annotation results of the MGEs, 46 subtypes were identified, most of which were transposases (Fig. S9). The MGE numbers among the different treatments did not show significant variations ($p <$

0.05, Fig. S10). However, the total abundance of MGEs was lower in HD and HG groups, although there was no significant difference compared to that in the MT and HT groups ($p < 0.05$, Fig. 3a). Notably, HB treatment significantly increased MGE abundance compared to that in the other TCMs addition groups ($p < 0.05$, Fig. 3a). In addition, variations in the dominant MGE types (transposases) and subtypes (*tnpA*) mirrored the total MGE abundance trend (Fig. S11).

The mobile ARGs were further identified based on the co-occurrence of ARG and MGE in the same contig. The results showed a lower proportion of mobile ARGs in the HD treatment than in the MT and HB treatments ($p < 0.05$; Fig. 3b). Some tetracycline-resistance genes were identified, including *tet(A)*, *tet(C)*, *tet(32)*, and *tet(40)* (Fig. 3c). Furthermore, the ranking of ARGs based on their human health risk was investigated. The abundance of Rank I was lowest in the HD and HG groups, which was markedly lower than that in the HT group ($p < 0.05$, Fig. 3d). Furthermore, the risk index calculated from the ARG ranks revealed the lowest values in the HD treatment group, which were markedly lower than those in MT treatment ($p < 0.05$; Fig. S12).

3.4. Host taxonomy of ARGs

As the most abundant ARG type, aminoglycoside resistance genes in the cecal microbiome of Wuding chickens were predominantly hosted by Firmicutes (Fig. 4a and b). Firmicutes were also the primary host for MLS, tetracycline, and streptothricin resistance genes (Fig. 4a and b). Additionally, Bacteroidetes was the main host for beta-lactam and MLS resistance genes, whereas Proteobacteria was another host for tetracycline resistance genes (Fig. 4a and b). Firmicutes were also the dominant host for other relatively rare ARG types, except for kasugamycin, multidrug, pepite, and sulfonamide resistance genes, which were predominantly carried by Proteobacteria (Fig. 4c). For ARG hosts in the different treatments, the results were almost identical between them, and consistent with the findings from all samples (Fig. S13). For a single

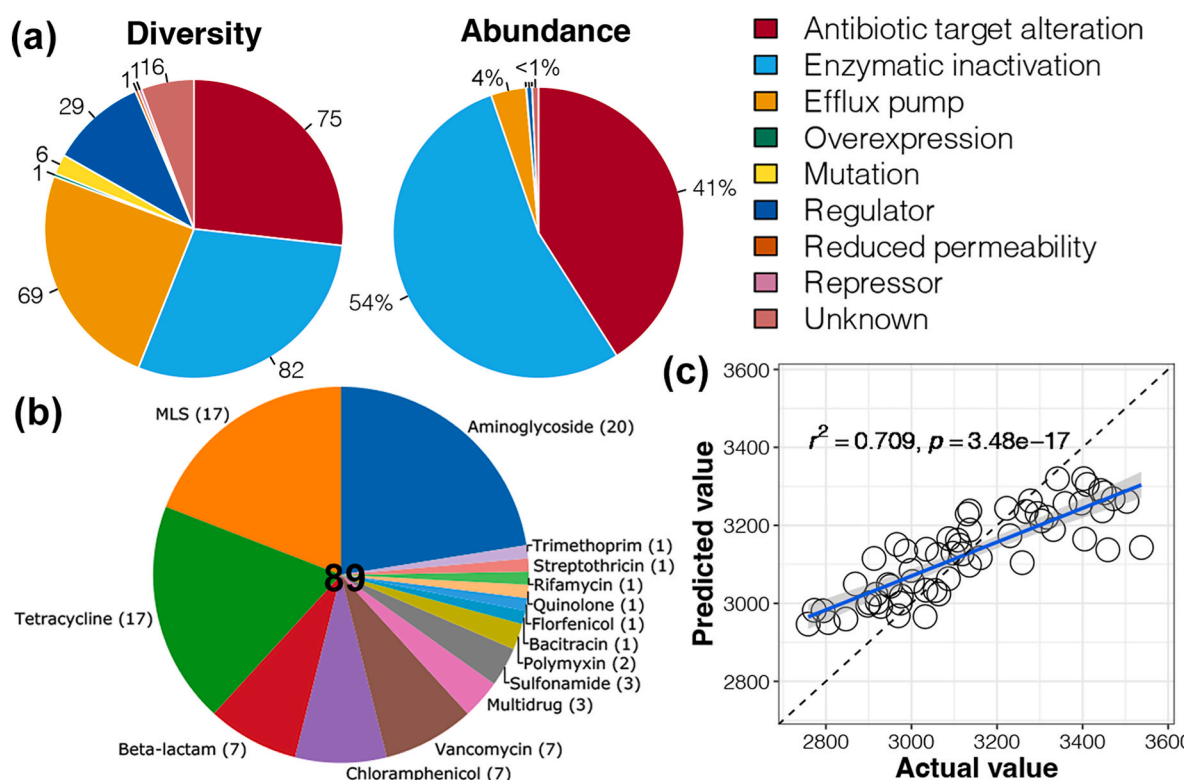


Fig. 2. Variations in resistance mechanisms and core ARG identification in cecal microbiome of Wuding chickens with different treatments. (a) Diversity and abundance proportion of different resistance mechanisms in our studied samples. (b) Distribution of shared ARGs at the type level. (c) Prediction efficiency of the total abundance of ARGs for random forest model based on the shared ARGs. R^2 and p -value in this plot are obtained by the linear regression.

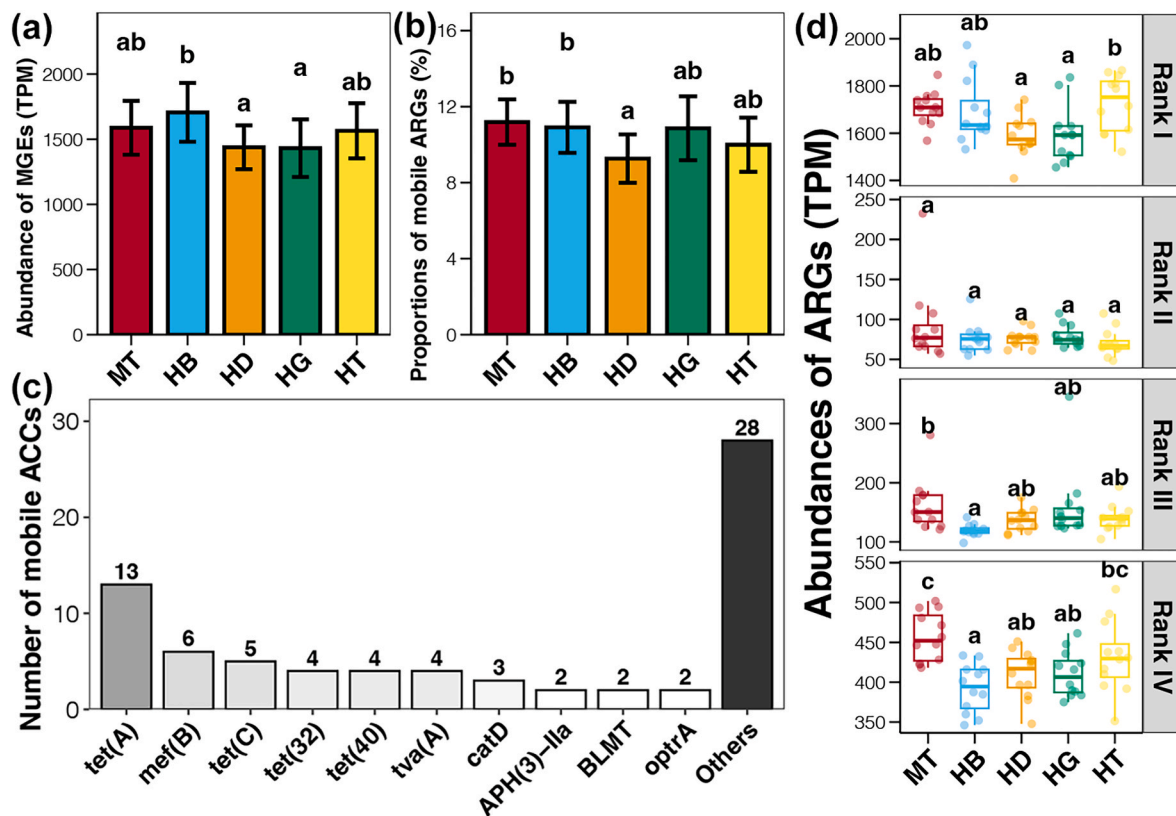


Fig. 3. Variations in mobilome and ARG risk in cecal microbiome of Wuding chickens with different treatments. (a) Differences in the total abundances of MGEs among different treatments. (b) Differences in the proportions of mobile ARGs in total ARGs among different treatments. (c) Number of mobile ACCs carrying various ARG subtypes. (d) Differences in abundance of different ARG ranks among different treatments. Different lowercases letters in each subfigure represent significant differences between different treatments (Tukey's HSD test, $p < 0.05$).

ARG, the top five most abundant ARG subtypes showed obvious variations in their abundance among different treatments, but not for their hosts, which were almost identical among the different treatments (Fig. S14).

3.5. PARB profiles

Procrustes alignment showed a marked goodness of fit between the antibiotic resistome and cecal bacterial composition ($p < 0.05$; Fig. 5a). In contrast, a lack of correlation between the ARG profile and the population of potential pathogens was observed (Fig. 5b). Only 45 pathogens carrying ARGs were identified in the dataset of this study, most of which exhibited no significant variations in abundance among the different treatments (Fig. 5c). However, certain PARB were more abundant in the MT samples, including *Enterocloster clostridioformis*, *Providencia rettgeri*, *Providencia stuartii*, *Serratia marcescens*, and *Serratia ureilytica* (Fig. 5c). Several PARB were enriched in the HB group, including *Bacteroides fragilis*, *Odoribacter laneus*, and *Parabacteroides distasonis* (Fig. 5c). Notably, the abundance of ESKAPE alone did not vary among the different treatments (Fig. 5c); however, the total abundance of ESKAPE was higher in the broiler diet group ($p < 0.05$, Fig. 5d). ARGs carried by PARB were further accounted for, and MRG resistance genes were revealed to be the most diverse ARG type in PARB, with most of them carried by *Escherichia coli* (*E. coli*) (Fig. 5d). Three contigs belonging to *E. coli* carrying multiple MRGs were found, all of which were obtained from HB samples (Fig. 5e).

3.6. Mechanisms for variations in antibiotic resistome

First, the correlations among the detected ARG numbers, MGE numbers, and cecal bacterial richness were investigated for each

treatment. Significant associations were observed between the number of ARGs and MGEs in four of the five treatments (linear regression, $p < 0.05$), except for the HD group (Fig. 6a). However, the correlation between ARG number and cecal bacterial richness was only significant in the HD treatment group (linear regression, $p < 0.05$; Fig. 6a). Associations between the antibiotic resistome and the mobilome or cecal bacterial community were further analyzed for each treatment, and significant correlations were observed in all measurements (linear regression, $p < 0.05$, Fig. 6b). Similarly, the strength of the correlation between the antibiotic resistome and mobilome was lowest in the HD treatment group, whereas that between the antibiotic resistome and cecal bacterial community was highest in the HD group (Fig. 6b). Additionally, stronger correlations were observed in all the above measurements in HT treatment than in MT treatment (Fig. 6a and b).

ARGs, MGEs, and cecal bacteria were categorized into several clusters based on their associations (Fig. 6c). MRG resistance genes were correlated with integrase, IS, and Proteobacteria. Bacteroidetes and transposases were correlated with MLS, tetracycline, and beta-lactam resistance genes. Vancomycin and streptothricin resistance genes were correlated with Firmicutes and Actinobacteria. Finally, variations in the relative importance of MGEs and cecal bacterial communities on the antibiotic resistome among the different treatments were explored (Fig. 6d). Compared with the MT group, the contribution of the interaction between MGEs and the cecal bacterial community to the antibiotic resistome increased in all high stocking density treatments. Moreover, the relative importance of the cecal bacterial community was elevated in the HD group, whereas the contribution of MGEs was reduced.

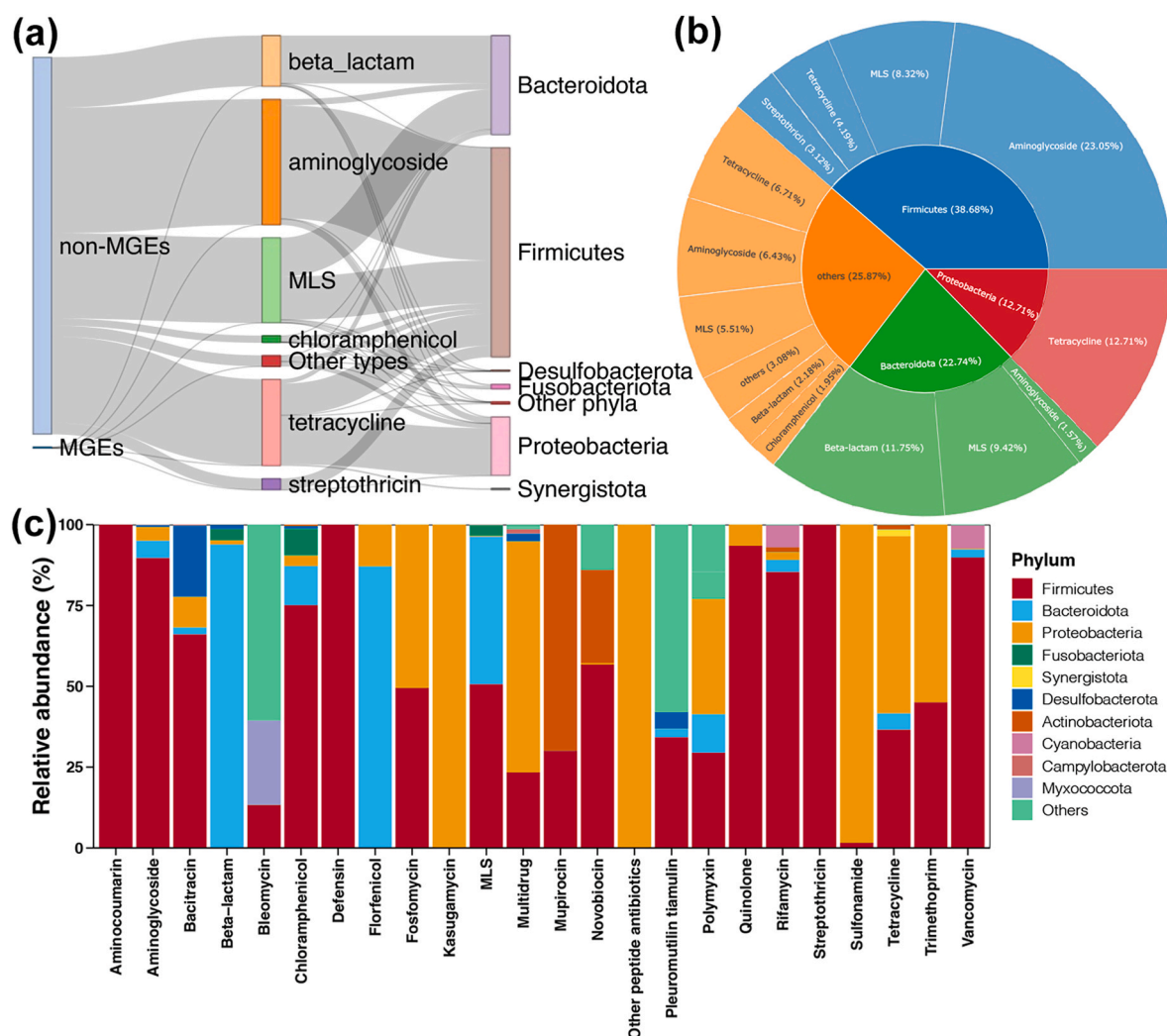


Fig. 4. Summary of the ARG hosts in cecal microbiome of Wuding chickens with different treatments. (a) Sankey diagrams indicating the co-existence of total ARGs with potential harbors. (b) Percentage of potential ARG hosts with different ARG types. (c) Barplot showing the compositions of hosts for different ARG types.

4. Discussion

4.1. Negligible impact of high stocking density on ARGs of Wuding chickens without using antibiotics

In this study, a breeding and feeding trial of Wuding chickens under one moderate stocking density mode and four high stocking density modes, supplemented with three different TCMs, were conducted. The first purpose was to detect the influence of high-density breeding on ARGs in Wuding chickens in the present research. Intensive breeding of livestock and poultry has resulted in an increased diversity of ARGs and greater resistance phenotypes in their feces, highlighting the impact of stocking density on ARG occurrence in animal farming (Wang et al., 2023). Similar findings have been reported in aquaculture, linking high stocking densities to a higher risk of pathogen outbreaks and ARG enrichment in the environment (Watts et al., 2017). In addition, increased stocking densities depresses chicken growth performance and affects digestive microbiota, potentially influencing the prevalence of ARGs (Guardia et al., 2011). However, in this study, no significant change in the diversity and abundance of ARGs and MGEs or in the resistance mechanisms, mobile ARGs, and ARG risks between the MT and HT groups were observed (Figs. 1a, 2b and 3). The only negative effect on ARGs was that high stocking density increased the individual difference levels of the antibiotic resistome in Wuding chickens (Fig. 1c).

The excellent characteristics of Wuding chickens as a high-quality breed may explain this phenomenon. Nevertheless, the strict prohibition of antibiotic use during the experimental process of this study could also play a pivotal role in controlling antibiotic resistance. Generally, the more intensive the aquaculture, the higher the amount of antibiotics used, which increases the risk of infection and makes animal husbandry a hotspot for ARGs (Zalewska et al., 2021). Thus, these results emphasize the importance of controlling antibiotic abuse for promoting sustainable and responsible animal husbandry practices.

4.2. Dietary berberine hydrochloride increasing the transmission potential of ARGs

Another purpose was to detect whether the addition of TCMs contributes to the control of antibiotic resistance in Wuding chicken farming. Differential responses of ARGs and MGEs to different TCMs were observed in the ceca of Wuding chickens (Figs. 1 and 3). Compared to treatments without TCMs, dietary berberine hydrochloride changed the ARG profiles (Fig. 1b). However, an increased abundance of MGEs and ESKAPE in the HB treatments was found, especially compared to the other TCM addition groups (Figs. 3a and 5d). HGT plays a crucial role in ARG spread among bacteria (Vinayamohan et al., 2022). This transfer of ARGs can lead to the development of highly pathogenic superbacteria that are resistant to most antimicrobial drugs, thus posing a serious

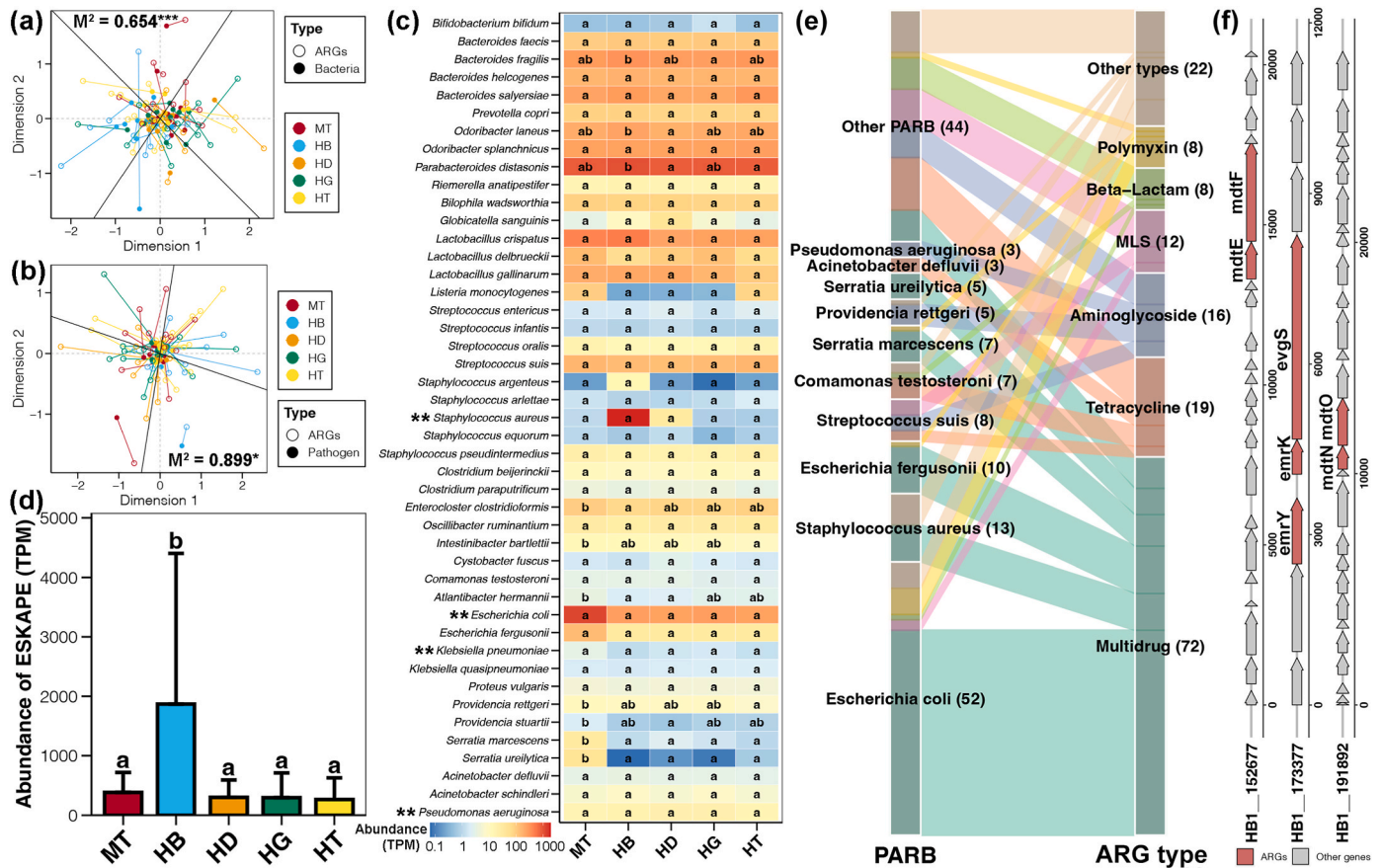


Fig. 5. Broad spectrum of PARB in cecal microbiome of Wuding chickens with different treatments. (a) Procrustes analysis showing the correlations of antibiotic resistome with total cecal bacteria. (b) Procrustes analysis showing the correlations of antibiotic resistome with potential pathogens. (c) Differences in the abundance of PARB among different treatments. (d) Differences in the total abundance of ESKAPE pathogens among different treatments. Different lowercases letters in each subfigure represent significant differences between different treatments (Tukey's HSD test, $p < 0.05$). (e) Sankey plot revealing the mapping relationship between PARB and ARGs across all samples. (f) Representative arrangements of *Escherichia coli* contigs carrying more than one multidrug resistance genes.

threat to human health (Tao et al., 2022). Additionally, harmless bacteria that acquire antibiotic resistance genes can transmit these genes to harmful bacteria, indirectly contributing to the spread of antibiotic resistance (Lerminiaux and Cameron, 2019). As HGT vehicles, the higher abundance of MGEs in the HB samples (Fig. 3a) indicated an increased transmission potential of ARGs under dietary berberine hydrochloride. Furthermore, the enriched ESKAPE carried ARGs (Fig. 5d) that further amplified the potential human health risks associated with the products and wastes from Wuding chickens fed berberine hydrochloride.

Notably, dietary berberine hydrochloride induced accumulation of multidrug-resistant PARB in the ceca of Wuding chickens (Fig. 5e). Among ARGs, MRGs are the most significant because of its role in conferring resistance to a wide range of antimicrobial agents (Van Duin and Paterson, 2016). This phenomenon is a major public health concern because it complicates treatment, increases the spread of resistance, and poses a serious global threat (Zhang et al., 2022). Bacteria carrying MRGs can lead to it being resistant to many antibiotics at once, making them difficult to treat and increasing healthcare costs (Nikaido, 2009). Despite detecting diverse MRGs (Fig. S1), their abundance was very low (Fig. S2). These results are inconsistent with the high abundance of MRGs in other ARG hotspots (Zhao, 2021b; Zhou et al., 2022). However, many of the low-abundance MRGs were present in the PARB genome (Fig. 5d), especially in *E. coli* in samples treated with berberine hydrochloride (Fig. 5e). As a highly regarded commensalism in the human gut, *E. coli*-carrying MRGs have been identified in animals, veterinary hospitals, and food animals, highlighting the importance of a single health approach to address antimicrobial resistance (Szmolka and Nagy, 2013).

More importantly, the presence of multidrug-resistant *E. coli* in both clinical and commensal settings underscores the interconnectedness of resistance mechanisms between human and animal populations and their potential for zoonotic transmission (Walker et al., 2022). The significance of MRGs is further underscored by the fact that they can be transmitted among bacteria, leading to the emergence of extensively drug-resistant and pan drug-resistant strains that are non-susceptible to multiple antimicrobial categories (Catalano et al., 2022). Fortunately, contig that simultaneously carried either MGEs or MRGs was not found (Fig. 5f). However, this did not change the fact that dietary berberine hydrochloride may have led to the spread of multidrug-resistant PARB during Wuding chicken farming.

4.3. Dietary dihydroartemisinin reduced ARGs through regulating the cecal microbiome of Wuding chickens

Among the five treatments in this study, the inclusion of dietary dihydroartemisinin resulted in the lowest number and abundance of ARGs in the ceca of Wuding chickens (Fig. 1a). Many ARG types and subtypes, as well as all resistance mechanisms, were observed, with the lowest abundance in the HD treatment group (Fig. 1d–S2, and S5). The most important reason for studying ARGs is that they have potential human health risks (Amarasiri et al., 2020). However, not all ARGs in non-clinical settings pose the same level of risk to human health (Gwenzi et al., 2020). Factors, such as clinical availability and human mobility, are important in evaluating the health risks associated with ARGs (Zhang et al., 2022). To pose substantial human health risks, ARGs need to transfer themselves to pathogens or hosts that are pathogenic (Ben

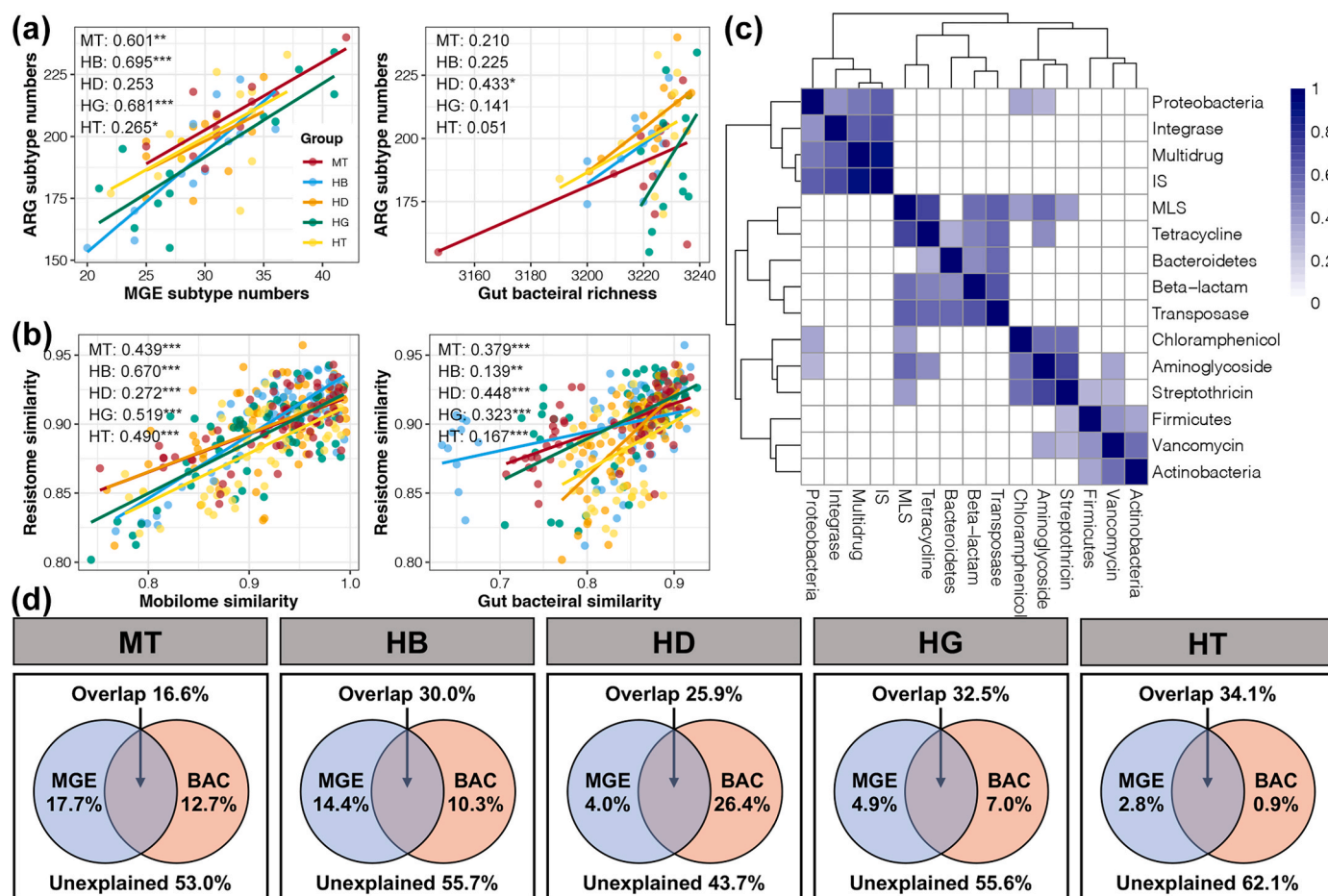


Fig. 6. Relative importances of MGEs and cecal bacterial community on variations in antibiotic resistome. (a) Linear regression of ARG numbers with the MGE numbers or cecal bacterial richness in each treatment, respectively. (b) Linear regression of antibiotic resistome with the mobilome or cecal bacterial community in each treatment, respectively. *, **, and *** represent the p-value of linear regression lower than 0.05, 0.01, and 0.001, respectively. (c) Heatmap revealing the relationships between abundances of ARG types, MGE types, and bacterial phyla. Color of block represent the correlation coefficient (Spearman correlation) and white color represent the p-value higher than 0.05. (d) VPA displaying the contribution of MGEs and cecal bacterial community on variations in antibiotic resistome of each treatment.

et al., 2019). In contrast to ARGs playing a “weapon-shield” role in clinical settings, it is quite not the circumstance in the natural environments (Martínez, 2008). Most ARGs are integrated into regulatory and metabolic networks in their original hosts (Piddock, 2006). For example, the primary function of the MFS family, a class of bacterial efflux pumps, is the uptake of sugars. However, Marger and Saier (1993) suggested that antibiotics and oligosaccharides are transported by MFS family member. The *acrB* gene, which encodes a multidrug efflux pump, is a global transcriptional regulator of carbohydrate catabolism (Perrenoud and Sauer, 2001). Therefore, when determining the risk of ARGs, it is necessary to differentiate them based on their risk factors (Bengtsson-Palme and Larsson, 2015). As agents with higher human health risks, the abundance of MGEs and the proportion of mobile ARGs were found to be the lowest in Wuding chickens fed dihydroartemisinin (Fig. 3a and b). Moreover, rank I ARGs with higher human health risk and the risk index based on ARG risk also ranked lowest in the HD samples (Fig. 3e and S12). These findings underscore the significant impact of dietary dihydroartemisinin on mitigating potential human health risks associated with antibiotic resistance during Wuding chicken farming.

Besides HGT, ARG development can also be influenced by vertical gene transfer (VGT) (Tao et al., 2022). VGT involves the transmission of genes from parents to offspring during bacterial division (Li et al., 2019a, b). Su et al. (2015) and Chen et al. (2016) suggested that the microbial community phylogeny is a critical factor in antibiotic

resistome aviation. While HGT remains a major driver in the evolution of untreatable “superbugs,” both VGT and HGT contribute significantly to the rapid spread of antibiotic resistance in the environment (Wang et al., 2021; Woods et al., 2020). However, VGT and HGT mechanisms shaping the antibiotic resistome vary (Wei et al., 2021), and understanding the dynamics of gene transfer mechanisms is crucial for developing strategies to control the spread of antibiotic resistance. A substantial increase in the contribution of VGT to the antibiotic resistome was observed, while the contribution of HGT was significantly reduced in samples fed dihydroartemisinin compared to other treatments (Fig. 6). The almost completely unchanged host composition of ARGs among treatments (Fig. 4) also suggests the limited potential of HGT for variations in the antibiotic resistome during Wuding chicken farming. The use of TCMs to combat ARB is expected to replace antibiotics and eliminate ARG-carrying pathogens. The findings of this study indicated that dietary dihydroartemisinin for Wuding chickens could reduce the risk of antibiotic resistance by effectively restricting HGT.

4.4. Implications and limitations

Long-term abuse of antibiotics and intensive breeding mode have made animal husbandry as a reservoir and transmission hotspot for ARGs. Currently, antibiotic alternatives have been frequently used in animal husbandry. TCMs have antibacterial properties, thus they may achieve the goal of inhibiting disease outbreaks during animal

husbandry similar to antibiotics. In this study, we compared the effects of three TCMs as dietary additives on antibiotic resistance risks in breeding poultry. Although one of the studied TCM showed negative effects on ARGs, other TCMs were found to promote the propagation and dissemination of ARGs, the most serious human health risks brought by antibiotic abuse. Thus, TCMs as antibiotic alternatives used in animal husbandry should be considered as “potential hazardous material” before conducting relevant research. This study provides metagenomic evidence of the development of ARGs in broiler breeding environments, highlighting the distinct characteristics of TCMs as feed additives. Based on the results of this study, in the future, when using alternative antibiotics in the animal husbandry industry, it is necessary to first confirm whether they have a positive effect on limiting the dissemination of ARGs. In addition, we should further study the potential risks of TCMs in other aspects, in order to gain a more comprehensive understanding of its application consequences and develop effective usage strategies.

5. Conclusions

High stocking density did not enrich ARGs in the ceca of Wuding chickens compared to moderate stocking density. Higher proportions of MGEs and some multidrug-resistant PARB were observed in Wuding chickens fed berberine hydrochloride than in chickens fed other TCMs, which could facilitate the HGT of ARGs and pose potential human health risks. In contrast, dietary dihydroartemisinin showed positive effects on the risk control of antibiotic resistance during Wuding chicken farming by reducing HGT and PARB. Although further investigations are needed to confirm and extend our findings, the use of dihydroartemisinin instead of antibiotics could effectively control the spread of ARGs in the intensive breeding of broilers.

CRedit authorship contribution statement

Hong Hu: Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ying Huang:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Changyou Shi:** Writing – review & editing, Formal analysis. **Anjian Li:** Formal analysis. **Qianhui Mi:** Formal analysis. **Kunping Wang:** Formal analysis. **Zelong Zhao:** Writing – original draft, Visualization, Software, Project administration. **Xi Bai:** Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hongbin Pan:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2024.143152>.

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